

Data Dictionary

Table 1: students

Database: student_db

Description: Stores one record per student with personal details.

Column	Data Type	Field Size	Constraint	Nullable	Java Mapping	Description
student_id	SERIAL (int)	4	PRIMARY KEY, Auto-increment	NO	int	Unique identifier for each student. Auto-generated by PostgreSQL. Never set manually on insert.
name	VARCHAR(100)	100	NOT NULL	NO	String	Full name of the student. Maximum 100 characters.

Table 2: subjects

Database: student_db

Description: Stores the list of subjects taught. Allows new subjects to be added without changing the table structure.

Column	Data Type	Field Size	Constraint	Nullable	Java Mapping	Description
subject_id	SERIAL (int)	4	PRIMARY KEY, Auto-increment	NO	int	Unique identifier for each subject. Auto-generated by PostgreSQL.
subject_name	VARCHAR(100)	100	NOT NULL, UNIQUE	NO	String	Name of the subject, e.g. Mathematics, English, Science, SST. UNIQUE ensures no duplicate subject names.

Table 3: scores

Database: student_db

Description: Links students to subjects and stores their score for each. This is the junction table that connects students and subjects.

Column	Data Type	Field Size	Constraint	Nullable	Java Mapping	Description
score_id	SERIAL (int)	4	PRIMARY KEY, Auto-increment	NO	int	Unique identifier for each score record. Auto-generated by PostgreSQL.
student_id	INT	4	NOT NULL, FOREIGN KEY -> students(student_id)	NO	int	References the student the score belongs to. If the student is deleted, their scores are deleted too (CASCADE).
subject_id	INT	4	NOT NULL, FOREIGN KEY -> subjects(subject_id)	NO	int	References the subject the score is for.

score	DOUBLE PRECISION	8	NOT NULL	NO	double	The student's score for t subject. 64-bit float, e.g. 78.5.
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Relationships Between Tables

students —< scores >— subjects

- One student can have many scores (one per subject).
- One subject can have many scores (one per student).
- scores is the junction table that sits between them.
- This is a many-to-many relationship resolved through scores.